

# M21 2026 July 2026 N250 Component

Joanna Morris

2026-08-11

## Contents

|          |                                                                      |           |
|----------|----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Load Files</b>                                                    | <b>2</b>  |
| <b>2</b> | <b>N250 WORD DATA: INTERACTION WITH OS (PRIMARY ANALYSES)</b>        | <b>3</b>  |
| 2.1      | N250 WORD DATA: FAMILY SIZE as Fixed Effect . . . . .                | 3         |
| 2.1.1    | Model . . . . .                                                      | 3         |
| 2.2      | N250 WORD DATA: BASE FREQUENCY as Fixed Effect . . . . .             | 5         |
| 2.2.1    | Model . . . . .                                                      | 5         |
| 2.3      | N250 WORD DATA: Combined Plots . . . . .                             | 6         |
| 2.4      | N250 WORD DATA: INTERACTION WITH LP (EXPLORATORY ANALYSES) . . . . . | 7         |
| 2.4.1    | Model . . . . .                                                      | 7         |
| 2.4.2    | Base Frequency x LP Interaction . . . . .                            | 7         |
| <b>3</b> | <b>NON-WORD DATA</b>                                                 | <b>10</b> |
| 3.1      | N250 NONWORD DATA: . . . . .                                         | 10        |
| 3.1.1    | Model . . . . .                                                      | 10        |
| 3.1.2    | Means and Contrasts . . . . .                                        | 11        |
| 3.1.3    | Plots . . . . .                                                      | 12        |
| 3.2      | N250 NONWORD DATA: OS and LP Interaction Plots . . . . .             | 12        |

Load libraries

Set emmeans options

```
emm_options(  
  lmer.df = "kenward-roger",  
  pbkrtest.limit = 10000,  
  disable.pbkrtest = FALSE  
)
```

Set ggplot parameters

```
# Color palette  
my_palette_1 <- c(  
  "#E31A1C", # red  
  "#1F78B4" # blue  
)  
  
my_palette_2 <- c(  
  "#0072B2", # dark blue  
  "#E69F00" # orange/gold  
)  
my_linetypes_1 <- c("dashed", "solid")  
  
# Theme  
my_theme <- theme_classic(base_size = 10) +  
  theme(  
    legend.position = "bottom",  
    legend.direction = "horizontal",  
    legend.box = "horizontal"  
  )  
  
# Common guides  
my_guides <- guides(  
  #
```

```

  colour = guide_legend(nrow = 1),
  fill   = guide_legend(nrow = 1)
)

```

Define 2-way interaction table function

# 1 Load Files

```

# Load data: rt words
rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item   = factor(target),
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>

```

```
mutate(
  SubjID = factor(SubjID),
  chlabel = factor(chlabel),
  base_frequency = factor(base_frequency,
    levels = c("Low", "High")),
  complexity = factor(complexity,
    labels = c("Simple", "Complex")),
  time_window = factor(time_window),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1)
)
```

## 2 N250 WORD DATA: INTERACTION WITH OS (PRIMARY ANALYSES)

At the N250, both the base frequency and family size models tested the interaction with Orthographic Sensitivity, consistent with the prediction that early form-based processing is selectively modulated by sublexical orthographic knowledge; Language Proficiency was included as a covariate in both models.

### 2.1 N250 WORD DATA: FAMILY SIZE as Fixed Effect

#### 2.1.1 Model

```
contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N250_fs <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_fs)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15026.8
||
|| Scaled residuals:
||    Min       1Q   Median       3Q      Max
|| -8.4669 -0.4045 -0.0218  0.3824 12.9276
||
|| Random effects:
|| Groups           Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  4.625      2.151
|| SubjID          (Intercept)  2.918      1.708
|| Residual                        2.754      1.659
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)   -0.75979    0.22921  57.00000   -3.315   0.0016 **
|| family_size1    0.32357    0.05837 1617.99999   5.543 3.46e-08 ***
|| OS              0.26085    0.25635  57.00000    1.018   0.3132
|| LP             -0.27860    0.18636  57.00000   -1.495   0.1404
|| family_size1:OS  0.11120    0.06532 1617.99999   1.702   0.0889 .
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 OS      LP
|| family_size1  0.000
|| OS            -0.046  0.000
|| LP             0.038  0.000  0.021
|| fmly_sz1:OS   0.000 -0.047  0.000  0.000
||
|| # Effect sizes for individual terms
eta_squared(m_w_N250_fs, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----
```

```

|| family_size |          0.02 | [0.01, 1.00]
|| OS          |          0.02 | [0.00, 1.00]
|| LP          |          0.04 | [0.00, 1.00]
|| family_size:OS |      1.79e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R2
r2_nakagawa(m_w_N250_fs)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.738
|| Marginal R2: 0.019

```

```

# family size
as.tibble(emmeans(m_w_N250_fs, ~ family_size)) |> print()

```

## Means and Contrasts

```

|| # A tibble: 2 x 6
||   family_size emmean   SE    df lower.CL upper.CL
||   <fct>       <dbl> <dbl> <dbl>   <dbl>   <dbl>
|| 1 Small      -0.900 0.231  58.9   -1.36   -0.438
|| 2 Large      -0.571 0.231  58.9   -1.03   -0.110

```

```

emm_fs_OS <- emmeans(m_w_N250_fs,
  ~ family_size | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

```

```

emm_part <- summary(emm_fs_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", family_size), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

```

```

contrast_part <- pairs(emm_fs_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

```

```

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  )

```

```

as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

```

```

|| # A tibble: 15 x 9
||   OS Effect          Estimate   SE    df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>          <dbl> <dbl> <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
|| 1 -1.9 Contrast: Small - Large    -0.11 0.14 1618      NA      NA    -0.8    0.421
|| 2 -1.9 EMM: Large      -1.19 0.55  58.9    -2.29 -0.08  NA     NA
|| 3 -1.9 EMM: Small     -1.3  0.55  58.9    -2.4  -0.19  NA     NA
|| 4 -1 Contrast: Small - Large    -0.21 0.09 1618      NA      NA   -2.37    0.018
|| 5 -1 EMM: Large      -0.9  0.35  58.9    -1.61 -0.19  NA     NA
|| 6 -1 EMM: Small     -1.11 0.35  58.9    -1.82 -0.4  NA     NA
|| 7 0 Contrast: Small - Large    -0.32 0.06 1618      NA      NA   -5.54    0
|| 8 0 EMM: Large      -0.58 0.23  58.9    -1.05 -0.12  NA     NA
|| 9 0 EMM: Small     -0.91 0.23  58.9    -1.37 -0.45  NA     NA
|| 10 1 Contrast: Small - Large    -0.43 0.09 1618      NA      NA   -5.08    0
|| 11 1 EMM: Large      -0.27 0.34  58.9    -0.95  0.41  NA     NA
|| 12 1 EMM: Small     -0.7  0.34  58.9    -1.38 -0.03  NA     NA
|| 13 1.9 Contrast: Small - Large    -0.53 0.13 1618      NA      NA   -3.97    0
|| 14 1.9 EMM: Large      0.02 0.53  58.9    -1.05  1.08  NA     NA
|| 15 1.9 EMM: Small    -0.52 0.53  58.9    -1.58  0.55  NA     NA

```

## Plots

## 2.2 N250 WORD DATA: BASE FREQUENCY as Fixed Effect

### 2.2.1 Model

```
contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N250_bf <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_bf)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 18999.1
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.4496 -0.3820 -0.0038  0.3905 14.0821
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept) 19.723    4.441
|| SubjID         (Intercept) 12.044    3.470
|| Residual                        8.095    2.845
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -1.5059    0.4650   56.9998  -3.239  0.00201 **
|| base_frequency1 -0.3183    0.1001 1617.9999  -3.181  0.00150 **
|| OS              0.5341    0.5200   56.9998   1.027  0.30874
|| LP             -0.5356    0.3781   56.9998  -1.417  0.16203
|| base_frequency1:OS -0.4549    0.1120 1617.9999  -4.062 5.09e-05 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) bs_fr1 OS      LP
|| bas_frqn1 0.000
|| OS        -0.046 0.000
|| LP         0.038 0.000 0.021
|| bs_frqn1:OS 0.000 -0.047 0.000 0.000
||
|| # Effect sizes for individual terms
|| eta_squared(m_w_N250_bf, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| base_frequency | 6.21e-03 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| LP | 0.03 | [0.00, 1.00]
|| base_frequency:OS | 0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R2
|| r2_nakagawa(m_w_N250_bf)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.801
|| Marginal R2: 0.018

as.tibble(emmeans(m_w_N250_bf, ~ base_frequency)) |> print()
```

### Means and contrasts

```
|| # A tibble: 2 x 6
||   base_frequency emmean    SE    df lower.CL upper.CL
||   <fct>          <dbl> <dbl> <dbl>   <dbl>   <dbl>
|| 1 Low          -1.29  0.467  58.3    -2.22   -0.355
```

```

|| 2 High          -1.63 0.467 58.3    -2.56 -0.692
emm_bf_OS <- emmeans(m_w_N250_bf,
  ~ base_frequency | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

emm_part <- summary(emm_bf_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE       = round(SE, 2),
    df       = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio  = round(t.ratio, 2),
    p.value  = round(p.value, 3)
  )

as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

```

```

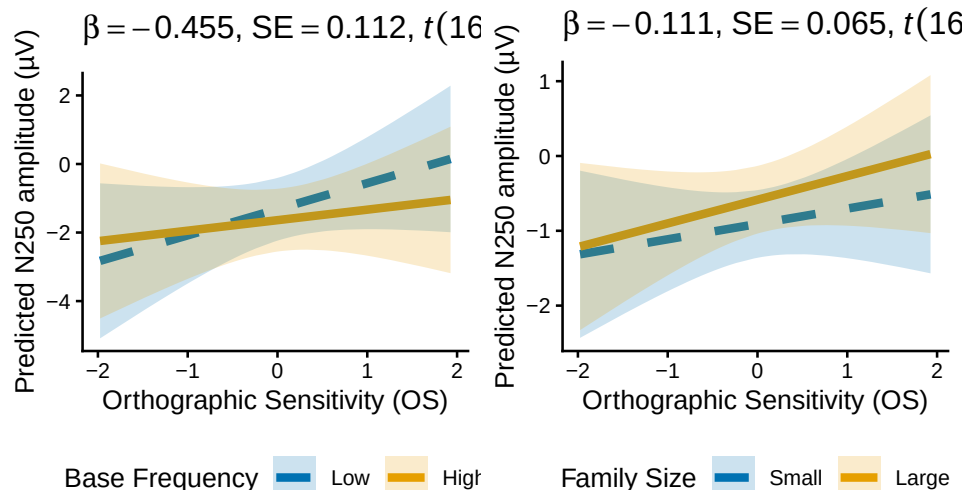
|| # A tibble: 15 x 9
||   OS Effect      Estimate SE      df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -1.9 Contrast: Low - High -0.55 0.24 1618    NA      NA    -2.28 0.023
|| 2 -1.9 EMM: High          -2.22 1.12 58.3   -4.46  0.02    NA    NA
|| 3 -1.9 EMM: Low          -2.77 1.12 58.3   -5.01 -0.53    NA    NA
|| 4 -1  Contrast: Low - High -0.14 0.15 1618    NA      NA    -0.89 0.374
|| 5 -1  EMM: High          -1.95 0.72 58.3   -3.38 -0.51    NA    NA
|| 6 -1  EMM: Low          -2.08 0.72 58.3   -3.52 -0.65    NA    NA
|| 7 0  Contrast: Low - High  0.32 0.1 1618    NA      NA     3.18 0.001
|| 8 0  EMM: High          -1.64 0.47 58.3   -2.57 -0.7     NA    NA
|| 9 0  EMM: Low          -1.32 0.47 58.3   -2.26 -0.39    NA    NA
||10 1  Contrast: Low - High  0.77 0.15 1618    NA      NA     5.27 0
||11 1  EMM: High          -1.33 0.68 58.3   -2.7  0.04    NA    NA
||12 1  EMM: Low          -0.56 0.68 58.3   -1.93 0.81    NA    NA
||13 1.9 Contrast: Low - High  1.18 0.23 1618    NA      NA     5.12 0
||14 1.9 EMM: High          -1.06 1.08 58.3   -3.21  1.1     NA    NA
||15 1.9 EMM: Low           0.13 1.08 58.3   -2.03  2.28    NA    NA

```

Plots

## 2.3 N250 WORD DATA: Combined Plots

### N250 Word Data



## 2.4 N250 WORD DATA: INTERACTION WITH LP (EXPLORATORY ANALYSES)

### 2.4.1 Model

```
# BINNED BY BASE FREQUENCY
m_w_N250_bf_LP <- lmer(
  value ~ base_frequency * LP + OS +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_bf_LP)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 19011.3
||
|| Scaled residuals:
||   Min       1Q   Median       3Q      Max
||  -7.3010  -0.3858  -0.0022   0.3832  14.2336
||
|| Random effects:
||   Groups             Name             Variance Std.Dev.
||   SubjID:chlabel    (Intercept)    19.694     4.438
||   SubjID             (Intercept)    12.044     3.470
||   Residual                        8.153     2.855
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -1.5059     0.4650   57.0001  -3.239  0.00201 **
|| base_frequency1 -0.3287     0.1004  1617.9999  -3.274  0.00108 **
|| LP             -0.5356     0.3780   57.0001  -1.417  0.16203
|| OS              0.5341     0.5200   57.0001   1.027  0.30874
|| base_frequency1:LP  0.1797     0.0817  1617.9999   2.200  0.02797 *
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) bs_fr1 LP      OS
|| bas_frqn1  0.000
|| LP          0.038  0.000
|| OS         -0.046  0.000  0.021
|| bs_frqn1:LP 0.000  0.039  0.000  0.000
||
|| # Effect sizes for individual terms
eta_squared(m_w_N250_bf_LP, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| base_frequency | 6.58e-03 | [0.00, 1.00]
|| LP | 0.03 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| base_frequency:LP | 2.98e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
r2_nakagawa(m_w_N250_bf_LP)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.799
|| Marginal R2: 0.018
```

### 2.4.2 Base Frequency x LP Interaction

```
as.tibble(emmeans(m_w_N250_bf_LP, ~ base_frequency)) |> print()

|| # A tibble: 2 x 6
||   base_frequency emmean    SE    df lower.CL upper.CL
||   <fct>          <dbl> <dbl> <dbl>   <dbl>   <dbl>
|| 1 Low          -1.29  0.467  58.3    -2.22   -0.355
|| 2 High          -1.63  0.467  58.3    -2.56   -0.692
```

```

emm_bf_LP <- emmeans(m_w_N250_bf_LP,
  ~ base_frequency | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3.0)),
  lmerTest.limit = 6480)

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect) |>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )

as.tibble(emm_contrast_table, digits = 3) |> print(n = 20)

|| # A tibble: 15 x 9
||   LP Effect      Estimate SE      df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -2.5 Contrast: Low - High 0.78 0.22 1618 NA NA 3.47 0.001
|| 2 -2.5 EMM: High -0.53 1.04 58.3 -2.62 1.55 NA NA
|| 3 -2.5 EMM: Low 0.24 1.04 58.3 -1.84 2.33 NA NA
|| 4 -1 Contrast: Low - High 0.51 0.13 1618 NA NA 4.01 0
|| 5 -1 EMM: High -1.2 0.59 58.3 -2.38 -0.02 NA NA
|| 6 -1 EMM: Low -0.69 0.59 58.3 -1.88 0.49 NA NA
|| 7 0 Contrast: Low - High 0.33 0.1 1618 NA NA 3.27 0.001
|| 8 0 EMM: High -1.65 0.47 58.3 -2.58 -0.71 NA NA
|| 9 0 EMM: Low -1.32 0.47 58.3 -2.25 -0.38 NA NA
|| 10 1 Contrast: Low - High 0.15 0.13 1618 NA NA 1.13 0.259
|| 11 1 EMM: High -2.09 0.61 58.3 -3.32 -0.86 NA NA
|| 12 1 EMM: Low -1.94 0.61 58.3 -3.17 -0.72 NA NA
|| 13 3 Contrast: Low - High -0.21 0.27 1618 NA NA -0.78 0.433
|| 14 3 EMM: High -2.98 1.25 58.3 -5.49 -0.48 NA NA
|| 15 3 EMM: Low -3.2 1.25 58.3 -5.7 -0.69 NA NA

# BINNED BY FAMILY SIZE

m_w_N250_fs_LP <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250_fs_LP)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15027.9
||
|| Scaled residuals:
||   Min      1Q   Median      3Q      Max
|| -8.4955 -0.3966 -0.0250  0.3751 12.8641
||
|| Random effects:
||   Groups      Name      Variance Std.Dev.
|| SubjID:chlabel (Intercept) 4.625   2.151
|| SubjID        (Intercept) 2.918   1.708
|| Residual                2.755   1.660
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||               Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   -0.75979    0.22921   56.99987  -3.315   0.0016 **
|| family_size1    0.32464    0.05836 1618.00005   5.563 3.11e-08 ***

```

```

|| LP          -0.27860    0.18636    56.99987   -1.495    0.1404
|| OS          0.26085    0.25635    56.99987    1.018    0.3132
|| family_size1:LP -0.07412    0.04749 1618.00005   -1.561    0.1188
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||          (Intr) fmly_1 LP      OS
|| family_size1  0.000
|| LP            0.038  0.000
|| OS            -0.046  0.000  0.021
|| fmly_sz1:LP   0.000  0.039  0.000  0.000
# Effect sizes for individual terms
eta_squared(m_w_N250_fs_LP, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| family_size    |          0.02 | [0.01, 1.00]
|| LP              |          0.04 | [0.00, 1.00]
|| OS              |          0.02 | [0.00, 1.00]
|| family_size:LP |       1.50e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
# Model R²
r2_nakagawa(m_w_N250_fs_LP)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.738
|| Marginal R2: 0.019

```

## 3 NON-WORD DATA

```
range(erp_fs_nw$LP)
```

```
|| [1] -2.674523  3.111265
```

```
range(erp_fs_nw$OS)
```

```
|| [1] -1.975820  1.926921
```

### 3.1 N250 NONWORD DATA:

For the NonWord data, BASE FREQUENCY and FAMILY SIZE are both COVARIATES. Because they are correlated only one needs to be included. Family Size is included as covariate, not fixed effect, because it explains more variance.

#### 3.1.1 Model

```
contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)
```

```
m_nw_N250_fs <- lmer(
  value ~ complexity * OS + complexity * LP + family_size +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250_fs)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * OS + complexity * LP + family_size + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 29933.2
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.7366 -0.5107 -0.0056  0.4938 12.2676
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  3.109      1.763
|| SubjID          (Intercept)  2.163      1.471
|| Residual                        4.072      2.018
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)  -0.66509    0.19682   56.99997  -3.379 0.001317 **
|| complexity1  -0.23263    0.05023 4856.00002  -4.631 3.73e-06 ***
|| OS           0.29808    0.22013   56.99997   1.354 0.181051
|| LP          -0.20675    0.16003   56.99997  -1.292 0.201577
|| family_size1 0.18172    0.05014 4856.00002   3.625 0.000292 ***
|| complexity1:OS 0.39199    0.05618 4856.00002   6.978 3.40e-12 ***
|| complexity1:LP 0.35764    0.04084 4856.00002   8.758 < 2e-16 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 OS      LP      fmly_1 cm1:OS
|| complexity1  0.000
|| OS          -0.046  0.000
|| LP           0.038  0.000  0.021
|| family_size1 0.000  0.000  0.000  0.000
|| cmplxty1:OS  0.000 -0.046  0.000  0.000  0.000
|| cmplxty1:LP  0.000  0.038  0.000  0.000  0.000  0.021
||
|| # Effect sizes for individual terms
|| eta_squared(m_nw_N250_fs, partial = TRUE)
```

```
|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----
|| complexity      | 4.40e-03 | [0.00, 1.00]
|| OS              | 0.03 | [0.00, 1.00]
|| LP              | 0.03 | [0.00, 1.00]
|| family_size     | 2.70e-03 | [0.00, 1.00]
|| complexity:OS   | 9.93e-03 | [0.01, 1.00]
```

```
|| complexity:LP | 0.02 | [0.01, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_nw_N250_fs)
```

```
|| # R2 for Mixed Models
||
|| Conditional R2: 0.575
|| Marginal R2: 0.025
```

### 3.1.2 Means and Contrasts

```
## complexity
as.tibble(emmeans(m_nw_N250_fs, ~ complexity))
```

```
|| # A tibble: 2 x 6
|| complexity emmean SE df lower.CL upper.CL
|| <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Simple -0.526 0.198 58.9 -0.922 -0.130
|| 2 Complex -0.759 0.198 58.9 -1.16 -0.363
```

```
## complexity x LP
emm_fs_LP <- emmeans(m_nw_N250_fs,
  ~ complexity | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3.0)))

emm_part <- summary(emm_fs_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

contrast_part <- pairs(emm_fs_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table_fs_LP <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )
# Force RStudio to show more rows inline by using the as.tibble() function
as_tibble(emm_contrast_table_fs_LP) |> print(n = 50)
```

```
|| # A tibble: 15 x 9
|| LP Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| <dbl> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -2.5 Contrast: Simple - Complex 1.11 0.11 4856 NA NA 9.92 0
|| 2 -2.5 EMM: Complex -0.69 0.44 58.9 -1.58 0.19 NA NA
|| 3 -2.5 EMM: Simple 0.42 0.44 58.9 -0.47 1.3 NA NA
|| 4 -1 Contrast: Simple - Complex 0.57 0.06 4856 NA NA 9.05 0
|| 5 -1 EMM: Complex -0.73 0.25 58.9 -1.23 -0.23 NA NA
|| 6 -1 EMM: Simple -0.16 0.25 58.9 -0.66 0.34 NA NA
|| 7 0 Contrast: Simple - Complex 0.22 0.05 4856 NA NA 4.31 0
|| 8 0 EMM: Complex -0.76 0.2 58.9 -1.16 -0.36 NA NA
|| 9 0 EMM: Simple -0.54 0.2 58.9 -0.94 -0.15 NA NA
|| 10 1 Contrast: Simple - Complex -0.14 0.07 4856 NA NA -2.14 0.032
|| 11 1 EMM: Complex -0.79 0.26 58.9 -1.31 -0.27 NA NA
|| 12 1 EMM: Simple -0.93 0.26 58.9 -1.45 -0.41 NA NA
|| 13 3 Contrast: Simple - Complex -0.86 0.13 4856 NA NA -6.38 0
|| 14 3 EMM: Complex -0.84 0.53 58.9 -1.91 0.22 NA NA
|| 15 3 EMM: Simple -1.7 0.53 58.9 -2.76 -0.64 NA NA
```

```
## complexity x OS
emm_fs_OS <- emmeans(m_nw_N250_fs,
  ~ complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)))

emm_part <- summary(emm_fs_OS) |>
  as.data.frame() |>
  transmute(OS, Effect = paste("EMM:", complexity), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)
```

```

contrast_part <- pairs(emm_fs_OS) |>
  summary() |>
  as.data.frame() |>
  transmute(OS, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

emm_contrast_table_fs_OS <- bind_rows(emm_part, contrast_part) |>
  arrange(OS, Effect)|> # sorts rows in the combined table, first by column "OS", then by "Effect"
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )
# Force RStudio to show more rows inline by using as.tibble()
as_tibble(emm_contrast_table_fs_OS) |> print(n = 50)

```

```

|| # A tibble: 15 x 9
||   OS Effect      Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <dbl> <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -1.9 Contrast: Simple - Complex    0.99  0.12 4856      NA      NA      8.29    0
|| 2 -1.9 EMM: Complex    -1.72  0.47  58.9    -2.67  -0.77    NA     NA
|| 3 -1.9 EMM: Simple    -0.72  0.47  58.9    -1.67  0.22     NA     NA
|| 4 -1 Contrast: Simple - Complex    0.64  0.08 4856      NA      NA      8.33    0
|| 5 -1 EMM: Complex    -1.27  0.3  58.9    -1.88  -0.66    NA     NA
|| 6 -1 EMM: Simple    -0.63  0.3  58.9    -1.24  -0.02    NA     NA
|| 7 0 Contrast: Simple - Complex    0.25  0.05 4856      NA      NA      4.98    0
|| 8 0 EMM: Complex    -0.78  0.2  58.9    -1.18  -0.38    NA     NA
|| 9 0 EMM: Simple    -0.53  0.2  58.9    -0.93  -0.13    NA     NA
|| 10 1 Contrast: Simple - Complex   -0.14  0.07 4856      NA      NA     -1.93   0.053
|| 11 1 EMM: Complex    -0.29  0.29  58.9    -0.87  0.3      NA     NA
|| 12 1 EMM: Simple    -0.43  0.29  58.9    -1.01  0.15     NA     NA
|| 13 1.9 Contrast: Simple - Complex  -0.49  0.12 4856      NA      NA     -4.27    0
|| 14 1.9 EMM: Complex    0.16  0.46  58.9    -0.76  1.07     NA     NA
|| 15 1.9 EMM: Simple   -0.34  0.46  58.9    -1.25  0.58     NA     NA

```

### 3.1.3 Plots

## 3.2 N250 NONWORD DATA: OS and LP Interaction Plots

### N250 NonWord Data

